MAT216: Linear Algebra and Fourier Transformation

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Lecture-13

Find the Fourier series of $f(x) = \frac{1}{2}(\pi - x), [0, 2\pi]$ and show that,

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$$

Answer:

$$A_0 = 0$$

$$A_k = 0$$

$$B_k = \frac{1}{\pi} \int_0^{2\pi} \frac{1}{2}(\pi - x)\sin(kx)dx = \cdots = \frac{1}{k}$$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos\left(k\frac{x\pi}{L}\right) + \sum_{k=1}^{\infty} b_k \sin\left(k\frac{x\pi}{L}\right)$$

$$a_k = \frac{1}{L}\int_{-L}^{L} f(x)\cos(k\frac{x\pi}{L})dx$$
$$b_k = \frac{1}{L}\int_{-L}^{L} f(x)\sin(k\frac{x\pi}{L})dx$$
$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)dx$$

How odd and even function change the formula for the Fourier series?

When $f(x)$ is even

$$A_k = \frac{1}{L}\int_{-L}^{L} \underbrace{f(x)\cos\left(k\frac{x\pi}{L}\right)}_{even} dx$$

$$= \frac{2}{L}\int_{0}^{L} f(x)\cos\left(k\frac{x\pi}{L}\right) dx$$

When $f(x)$ is odd

$$B_k = \frac{1}{L}\int_{-L}^{L} \underbrace{f(x)\sin\left(k\frac{x\pi}{L}\right)}_{even} dx$$

$$= \frac{2}{L}\int_{0}^{L} f(x)\sin\left(k\frac{x\pi}{L}\right) dx$$

Find the Fourier Sine Series of $f(x) = \cos(x)$ on the interval $(0, \pi)$.

Perseval Identity:

$$\langle f(x), f(x) \rangle = \frac{A_0^2}{2} + \sum_{k=1}^{\infty} A_k^2 + B_k^2$$

Find the Fourier series of $f(x) = x$ on the interval $(-\pi, \pi)$ and show that,

$$\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

Answer: We already have,

$$f(x) = \sum_{k=1}^{\infty} -\frac{2(-1)^k}{k} \sin(kx)$$